Revisiting Glow Models for River Level Prediction

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2025

Abstract

We revisit glow models for predicting fictional river levels from a small corpus of gauge records. On the GlowRiver corpus and on RiverLight, a gradient boosting model reaches an accuracy of 0.78 and an RMSE of 1.60.

1 Introduction

Predicting river levels with few gauges is still difficult. The aim of this work is to test whether gradient boosting is enough when the corpus is small.

2 Method

We train a gradient boosting model with 300 trees and a learning rate of 0.05 on the GlowRiver corpus of 2,400 gauge records, and evaluate on RiverLight.

3 Results

The gradient boosting model reached an accuracy of 0.78 and an RMSE of 1.60 on RiverLight, lower than the values reported for lantern networks.

4 Limitations

A limitation is that the GlowRiver corpus covers only one invented season. Future work should add winter records.